

BS-CH101

Chemistry-I

Combined Module Notes

Modules 1 – 7

Semester I • 3L:1T:0P • 4 Credits

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Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Module 1

Atomic and Molecular Structure

1. Atomic and Molecular Structure

Atomic and molecular structure explains the arrangement of electrons in atoms and molecules, the origin of bonding, and the connection between energy levels and observable properties.

1.1 Schrodinger Wave Equation

In quantum chemistry, an electron is described by a wave function ψ . The square of the wave function gives probability density, so $|\psi|^2 d\tau$ is the probability of finding the electron in a small volume element $d\tau$.

Time-Independent Schrodinger Equation

$$\hat{H}\psi = E\psi$$

where $\hat{H}$ is the Hamiltonian operator, ψ is the wave function, and E is the permitted energy. For a one-dimensional particle,

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi$$

1.2 Particle in a One-Dimensional Box

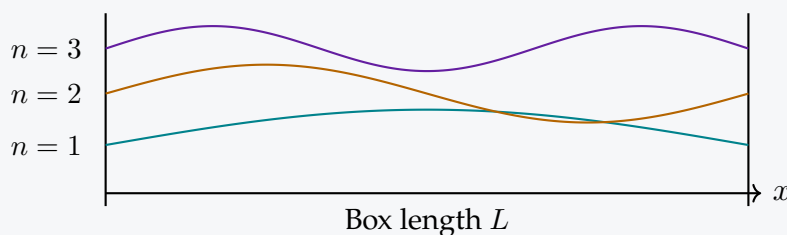
For a particle confined between $x = 0$ and $x = L$, the potential energy is zero inside the box and infinite outside it. Boundary conditions are:

$$\psi(0) = 0, \quad \psi(L) = 0$$

The allowed wave functions and energies are:

$$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad E_n = \frac{n^2 h^2}{8mL^2}, \quad n = 1, 2, 3, \dots$$

Particle in a Box: Quantized Standing Waves



1.3 Molecular Orbitals of Diatomic Molecules

Molecular orbital (MO) theory states that atomic orbitals combine to form molecular orbitals spread over the whole molecule. Constructive overlap gives bonding MOs and destructive overlap gives antibonding MOs.

Bond Order

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

where N_b is the number of electrons in bonding MOs and N_a is the number of electrons in antibonding MOs. Higher bond order means shorter, stronger bond.

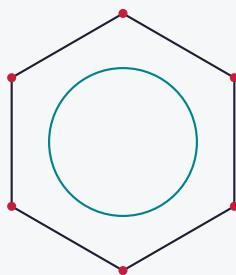
Important examples:

Molecule	Configuration idea	Bond order	Magnetism
H ₂	Two electrons in σ_{1s}	1	Diamagnetic
O ₂	Two unpaired electrons in π_{2p}^*	2	Paramagnetic
N ₂	Filled bonding orbitals, no antibonding excess	3	Diamagnetic

1.4 Pi-Molecular Orbitals and Aromaticity

Pi-electrons in conjugated systems are delocalized. Benzene has six pi-electrons distributed over six carbon atoms, making it unusually stable.

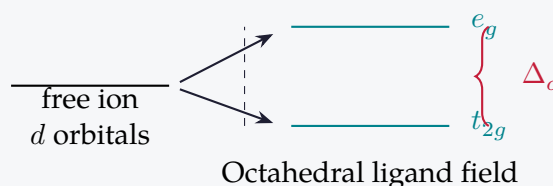
Aromatic compound: A cyclic, planar, fully conjugated molecule having $(4n + 2)$ pi-electrons, where $n = 0, 1, 2, \dots$

Benzene Pi-Electron Delocalization

6 delocalized pi-electrons satisfy Huckel rule

1.5 Crystal Field Theory

In transition-metal complexes, surrounding ligands split the five degenerate d orbitals into different energy groups. The splitting controls colour, magnetic behaviour, and stability.

Octahedral Crystal Field Splitting**1.6 Band Structure and Doping**

In solids, many atomic orbitals combine to form bands. Conductivity depends on the band gap E_g between valence band and conduction band.

Material	Band gap	Electrical behaviour
Conductor	Overlapping bands or very small gap	Electrons move easily
Semiconductor	Moderate gap, about 1–3 eV	Conductivity increases by heat or doping
Insulator	Large gap, usually > 5 eV	Very poor conductor

Exam Points

- Derive $E_n = n^2 h^2 / (8mL^2)$ using boundary conditions.
- For MO questions, always calculate bond order and then state magnetic nature.
- For CFT, mention splitting pattern, Δ_o , high spin or low spin, and magnetic moment.
- For band theory, draw valence band, conduction band, and band gap.

2. Exam-Oriented Derivations and Applications

2.1 Derivation of Particle in a Box Energy

For a one-dimensional box, potential energy inside the box is zero. Therefore the Schrodinger equation becomes:

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi}{dx^2} = E\psi$$

Rearranging,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2mE}{h^2}\psi = 0$$

Let

$$k^2 = \frac{8\pi^2mE}{h^2}$$

Then the differential equation becomes:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

Its general solution is:

$$\psi = A \sin kx + B \cos kx$$

Boundary condition at $x = 0$ gives $B = 0$. Boundary condition at $x = L$ gives:

$$A \sin kL = 0$$

For a non-zero wave function:

$$kL = n\pi, \quad k = \frac{n\pi}{L}$$

Substituting in $k^2 = 8\pi^2mE/h^2$:

$$E_n = \frac{n^2h^2}{8mL^2}$$

Solved Example: Energy Level Ratio

Problem: Find the ratio of the first three energy levels for a particle in a box.

Solution: Since $E_n \propto n^2$,

$$E_1 : E_2 : E_3 = 1^2 : 2^2 : 3^2 = 1 : 4 : 9$$

Exam note: Energy gaps are not equal; they increase with n .

2.2 MO Filling Order and Magnetic Behaviour

For second-period diatomic molecules, MO ordering changes after N_2 because s - p mixing becomes less important. This is a common exam trap.

Molecules	MO order to remember	Important result
B_2, C_2, N_2	$\sigma_{2s} < \sigma_{2s}^* < \pi_{2p} < \sigma_{2p} < \pi_{2p}^* < \sigma_{2p}^*$	B_2 is paramagnetic; N_2 has bond order 3
O_2, F_2, Ne_2	$\sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} < \pi_{2p}^* < \sigma_{2p}^*$	O_2 is paramagnetic due to two unpaired electrons

Solved Example: Bond Order of Oxygen

For O_2 , total valence electrons are 12.

$$N_b = 8, \quad N_a = 4$$

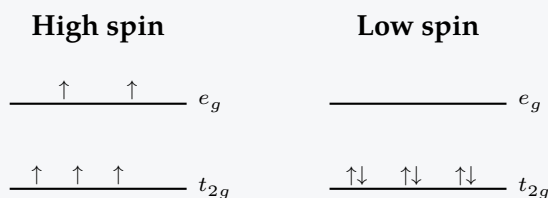
$$\text{Bond order} = \frac{8 - 4}{2} = 2$$

Two electrons remain unpaired in π_{2p}^* orbitals, so oxygen is **paramagnetic**.

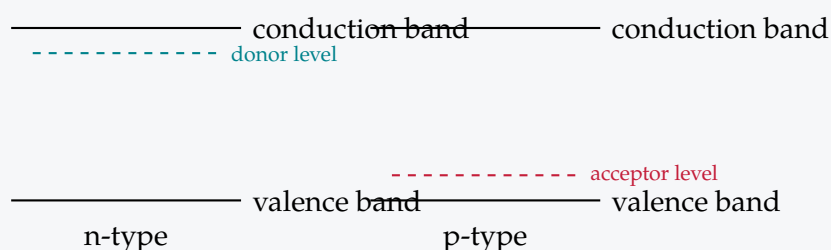
2.3 Crystal Field Splitting: High Spin and Low Spin

In octahedral complexes, electrons first occupy lower t_{2g} orbitals. Whether pairing occurs depends on the relative values of crystal field splitting energy Δ_o and pairing energy P .

Case	Condition	Result
High spin complex	$\Delta_o < P$	Electrons avoid pairing; more unpaired electrons; usually weak-field ligands
Low spin complex	$\Delta_o > P$	Electrons pair in t_{2g} before occupying e_g ; fewer unpaired electrons; usually strong-field ligands

High Spin and Low Spin Choice for Octahedral Complexes**2.4 n-Type and p-Type Semiconductors**

Doping adds controlled impurities to a semiconductor. Pentavalent dopants such as P or As introduce donor levels and produce n-type semiconductors. Trivalent dopants such as B or Al introduce acceptor levels and produce p-type semiconductors.

Doping Levels in Semiconductors

How to Write 10-Mark Answers in This Module

1. Start with a definition in 2–3 lines.
2. Draw the required energy-level or orbital diagram.
3. Write the governing formula with symbol meanings.
4. Add one example such as O₂ paramagnetism, benzene aromaticity, or n-type doping.
5. End with one application or exam conclusion.

2.5 Important Long-Answer: Aromaticity of Benzene

Benzene is the standard example of aromatic stability. A complete answer should not only state Huckel rule, but also connect structure, pi-electron count, planarity, and equal bond lengths.

1. Benzene is cyclic and planar, so all six carbon atoms can overlap through unhybridized *p* orbitals.
2. Each carbon contributes one pi-electron, giving a total of six pi-electrons.
3. Huckel rule requires $(4n + 2)$ pi-electrons. For benzene:

$$4n + 2 = 6 \quad \Rightarrow \quad n = 1$$
4. The pi-electrons are delocalized above and below the ring.
5. Delocalization makes all C–C bonds equivalent and gives extra stability.

Exam Comparison: Aromatic, Anti-Aromatic and Non-Aromatic

Type	Condition	Pi-electrons	Example idea
Aromatic	Cyclic, planar, conjugated	$(4n + 2)$	Benzene, cyclopentadienyl anion
Anti-aromatic	Cyclic, planar, conjugated	$4n$	Cyclobutadiene type systems
Non-aromatic	Fails planarity or conjugation	Rule not applicable	Non-planar or saturated ring

2.6 Important Long-Answer: Crystal Field Theory and Magnetism

Crystal field theory explains colour and magnetic properties of transition metal complexes by ligand-induced splitting of *d* orbitals.

Crystal Field Stabilization Energy

For an octahedral complex:

$$\text{CFSE} = (-0.4x + 0.6y)\Delta_o + pP$$

where *x* is the number of electrons in *t*_{2g} orbitals, *y* is the number of electrons in *e*_g orbitals, and *pP* is the pairing energy contribution.

Solved Example: Octahedral d^3 Configuration

For a d^3 octahedral ion, electrons occupy t_{2g} orbitals singly:



$$\text{CFSE} = 3(-0.4\Delta_o) = -1.2\Delta_o$$

There are three unpaired electrons, so the spin-only magnetic moment is:

$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$$

2.7 Band Theory: Exam Numericals and Concepts

Band gap decides electrical behaviour. If the gap is small, thermal energy can excite electrons to the conduction band. Doping reduces the effective energy needed for charge conduction.

Question type	What to write	Common conclusion
Conductor vs semiconductor	Draw overlapping band and small band gap	Conductor has many free electrons
Intrinsic semiconductor	Pure semiconductor with thermally generated carriers	Electron and hole concentrations are equal
n-type semiconductor	Donor level near conduction band	Electrons are majority carriers
p-type semiconductor	Acceptor level near valence band	Holes are majority carriers

Previous-Year Style Questions

1. Derive the energy expression for a particle in a one-dimensional box and explain why $n = 0$ is not allowed.
2. Draw the MO diagram of O_2 and explain its paramagnetic nature.
3. Explain aromaticity of benzene using Huckel rule.
4. Explain high-spin and low-spin octahedral complexes with suitable examples.
5. Compare conductors, semiconductors and insulators using band theory.

Quick Revision — Module 1

Atomic and Molecular Structure | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\hat{H}\psi = E\psi$	Schrodinger eigenvalue equation for allowed energies.
$E_n = \frac{n^2 h^2}{8mL^2}$	Energy of a particle in a one-dimensional box.
$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$	Normalized wave function for a particle in a box.
$\text{Bond order} = \frac{N_b - N_a}{2}$	Predicts bond strength and stability in MO theory.
$\mu = \sqrt{n(n+2)} \text{ BM}$	Spin-only magnetic moment for n unpaired electrons.
$4n + 2$	Huckel rule for aromatic pi-electron count.

One-Line Definitions

- **Wave function:** Mathematical function whose square gives probability density.
- **Node:** Point or surface where $\psi = 0$.
- **Bonding MO:** Lower-energy MO formed by constructive overlap.
- **Antibonding MO:** Higher-energy MO formed by destructive overlap.
- **Aromaticity:** Extra stability due to cyclic delocalization of $(4n + 2)$ pi-electrons.
- **Crystal field splitting:** Separation of metal d orbitals in a ligand field.
- **Band gap:** Energy gap between valence and conduction bands.

Frequently Asked / Viva Questions

Q1. State the Schrodinger equation.

The time-independent form is $\hat{H}\psi = E\psi$, where $\hat{H}$ is the energy operator.

Q2. Why is energy quantized for a particle in a box?

Only wave functions satisfying $\psi(0) = 0$ and $\psi(L) = 0$ are allowed, giving discrete n values.

Q3. What is the zero-point energy?

The minimum energy of a confined particle, $E_1 = h^2/(8mL^2)$.

Q4. Define bond order.

Bond order is $(N_b - N_a)/2$; it measures net bonding in a molecule.

Q5. Why is oxygen paramagnetic?

Frequently Asked / Viva Questions (contd.)

O₂ has two unpaired electrons in antibonding π^* orbitals.

Q6. What is Huckel rule?

A cyclic planar conjugated molecule is aromatic if it has $(4n + 2)$ pi-electrons.

Q7. Why is benzene stable?

Six pi-electrons are delocalized over the ring, satisfying Huckel rule with $n = 1$.

Q8. What is crystal field splitting?

Splitting of metal d orbitals into groups of different energy by ligands.

Q9. What is the role of doping in semiconductors?

Doping introduces donor or acceptor levels and increases carrier concentration.

Q10. How are conductors and insulators distinguished by band theory?

Conductors have overlapping or nearly touching bands, whereas insulators have a large band gap.

Summary Reference Table

Topic	Key idea	Exam output
Particle in box	Boundary conditions give discrete energies	Derivation plus energy-level diagram
MO theory	Atomic orbitals combine into bonding and antibonding MOs	Bond order and magnetism
CFT	Ligands split d orbitals	Splitting diagram and unpaired electrons
Band theory	Bands decide electrical conductivity	Conductor, semiconductor, insulator comparison

Module 2

Spectroscopic Techniques and Applications

1. Spectroscopic Techniques and Applications

Spectroscopy is the study of interaction between electromagnetic radiation and matter. It is used to identify molecules, determine structure, and study energy-level transitions.

1.1 Basic Principle and Selection Rules

When radiation of frequency ν is absorbed or emitted, a molecule moves between two energy levels.

Energy of Radiation

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

where h is Planck's constant, ν is frequency, c is speed of light, and λ is wavelength.

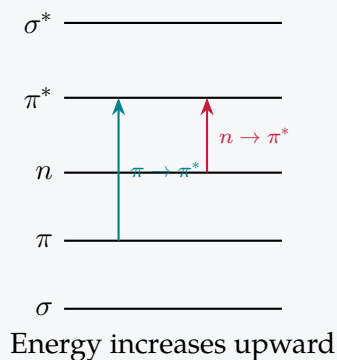
A transition is allowed only when it satisfies the required selection rule and has a changing electric or magnetic moment.

Technique	Energy change	Common selection rule
Rotational	Molecular rotation	$\Delta J = \pm 1$
Vibrational	Bond vibration	$\Delta v = \pm 1$ for fundamental transition
Electronic	Electron promotion	Spin and symmetry allowed transitions are intense
NMR	Nuclear spin orientation	Nuclei must have non-zero spin

1.2 Electronic Spectroscopy

Electronic spectroscopy uses UV-visible radiation. It involves transitions such as $\sigma \rightarrow \sigma^*$, $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, and $n \rightarrow \pi^*$.

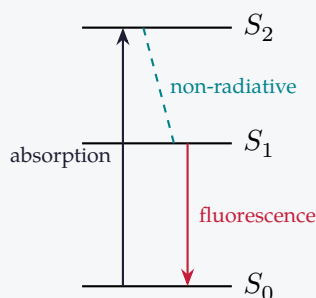
Common Electronic Transitions



1.3 Fluorescence and Medical Applications

Fluorescence occurs when a molecule absorbs light and quickly emits lower-energy light. It is useful in diagnostics, bio-imaging, and detection of trace biomolecules.

Jablonski Diagram for Fluorescence



1.4 Vibrational and Rotational Spectroscopy

IR spectroscopy studies molecular vibrations. A vibration is IR active if it changes dipole moment. Rotational spectroscopy is observed mainly for gas-phase polar molecules.

Vibrational Frequency of a Diatomic Molecule

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

where k is force constant and μ is reduced mass.

1.5 NMR, MRI, Surface Characterisation, Diffraction

NMR detects nuclei in a magnetic field. MRI is a medical imaging application of NMR, mainly using hydrogen nuclei in water and fat. Surface characterisation methods such as XPS, SEM, and AFM help study surface composition and morphology. Diffraction and scattering reveal crystal structure, particle size, and defects.

Bragg Equation

$$n\lambda = 2d \sin \theta$$

Here d is interplanar spacing and θ is the Bragg angle.

Exam Points

- Mention the energy region: microwave for rotational, IR for vibrational, UV-visible for electronic.
- In fluorescence answers, draw the Jablonski diagram.
- For NMR/MRI, write the need for non-zero nuclear spin.
- For X-ray diffraction, Bragg equation is compulsory.

2. Detailed Spectroscopy Notes for Exam Answers

2.1 Electromagnetic Spectrum and Type of Transition

Each spectroscopic method works in a particular energy region. In exam answers, first identify the radiation region, then the molecular transition.

Region	Approximate wavelength / frequency	Transition observed
Radiofrequency	MHz range	Nuclear spin transitions in NMR
Microwave	1 mm to 1 m	Rotational transitions
Infrared	4000–400 cm^{-1}	Vibrational transitions
Visible / UV	800–200 nm	Electronic transitions
X-ray	Very short wavelength	Diffraction by crystal planes

2.2 Beer-Lambert Law in Electronic Spectroscopy

UV-visible spectroscopy is often used for concentration measurement. If monochromatic light passes through a solution:

$$A = \log \frac{I_0}{I}$$

where I_0 is incident intensity and I is transmitted intensity.

Beer-Lambert Law

$$A = \epsilon cl$$

where A is absorbance, ϵ is molar absorptivity in $\text{L mol}^{-1} \text{cm}^{-1}$, c is concentration in mol L^{-1} , and l is path length in cm.

Solved Example: Absorbance

If $\varepsilon = 2.0 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$, $c = 1.0 \times 10^{-5} \text{ mol L}^{-1}$, and $l = 1 \text{ cm}$:

$$A = \varepsilon cl = (2.0 \times 10^4)(1.0 \times 10^{-5})(1) = 0.20$$

So the solution has absorbance 0.20.

2.3 IR Spectroscopy: Functional Group Region

IR spectra are very useful for identifying functional groups. The fingerprint region helps confirm identity, while the functional group region gives quick structural clues.

Bond / group	Typical range	Exam identification
O–H stretch	3200–3600 cm^{-1}	Broad band for alcohols and acids
N–H stretch	3300–3500 cm^{-1}	Amine or amide indication
C=O stretch	1650–1800 cm^{-1}	Strong sharp carbonyl peak
C=C stretch	1600–1680 cm^{-1}	Alkene or aromatic ring
C–O stretch	1000–1300 cm^{-1}	Alcohol, ether, ester

2.4 NMR Chemical Shift and Splitting

In proton NMR, chemical shift tells the electronic environment. Spin-spin splitting tells the number of neighbouring equivalent protons.

NMR Splitting Rule

If a proton has n equivalent neighbouring protons, its signal is split into:

$$n + 1 \text{ peaks}$$

The relative intensities approximately follow Pascal's triangle.

Proton type	Typical chemical shift	Reason
Alkyl R – CH ₃	0.8–1.5 ppm	Shielded environment
Near electronegative atom	3–4.5 ppm	Deshielding by O, N, halogen
Aromatic proton	6.5–8 ppm	Ring current effect
Aldehydic proton	9–10 ppm	Strongly deshielded

2.5 Bragg Law Derivation Idea

Constructive interference occurs when the path difference between X-rays reflected from successive crystal planes is an integral multiple of wavelength.

$$\text{Path difference} = AB + BC = 2d \sin \theta$$

For constructive interference:

$$n\lambda = 2d \sin \theta$$

This equation is used to calculate interplanar spacing d from diffraction angle.

Solved Example: Crystal Plane Spacing

For first-order reflection, $\lambda = 1.54 \text{ \AA}$ and $\theta = 30^\circ$:

$$d = \frac{n\lambda}{2 \sin \theta} = \frac{1.54}{2 \sin 30^\circ}$$

$$d = \frac{1.54}{1} = 1.54 \text{ \AA}$$

Answer-Writing Pattern for Spectroscopy

1. State the radiation region and energy transition.
2. Give the selection rule or activity condition.
3. Draw the transition diagram if possible.
4. Mention one analytical or medical application.
5. For numerical questions, write units clearly.

2.6 Rotational Spectroscopy of a Rigid Diatomic Molecule

For a rigid rotor, rotational energy levels are quantized:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1), \quad J = 0, 1, 2, \dots$$

where $I = \mu r^2$ is moment of inertia, μ is reduced mass, and r is bond length.

Rotational Spectral Lines

In wavenumber units:

$$\bar{\nu} = 2B(J+1), \quad B = \frac{h}{8\pi^2 I c}$$

Successive lines are equally spaced by:

$$\Delta \bar{\nu} = 2B$$

Solved Example: Rotational Line Spacing

If the rotational constant of a molecule is $B = 5 \text{ cm}^{-1}$, then adjacent rotational spectral lines are separated by:

$$\Delta \bar{\nu} = 2B = 10 \text{ cm}^{-1}$$

2.7 Vibrational Spectroscopy and Hooke's Law Model

A diatomic molecule behaves approximately like two masses connected by a spring. Stronger bonds have larger force constant k and therefore absorb at higher wavenumber.

Factor	Effect on frequency	Reason
Higher force constant	Frequency increases	Bond is stiffer
Higher reduced mass	Frequency decreases	Heavier atoms vibrate more slowly
Hydrogen bonding	O–H/N–H peak becomes broad	Different bond strengths in association
Conjugation near C=O	C=O frequency decreases slightly	Partial single-bond character increases

2.8 Electronic Spectra: Bathochromic and Hypsochromic Shifts

Electronic absorption changes when substituents or solvents change the energy gap between molecular orbitals.

Term	Meaning	Simple interpretation
Bathochromic shift	Shift to longer wavelength	Energy gap decreases; red shift
Hypsochromic shift	Shift to shorter wavelength	Energy gap increases; blue shift
Hyperchromic effect	Increase in absorption intensity	More allowed transition or greater conjugation
Hypochromic effect	Decrease in absorption intensity	Less allowed transition or restricted structure

2.9 Fluorescence Versus Phosphorescence

Fluorescence and phosphorescence are both emission processes, but their lifetimes and electronic states differ.

Point	Fluorescence	Phosphorescence
Transition	Singlet to singlet	Triplet to singlet
Lifetime	Very short, about 10^{-9} – 10^{-7} s	Longer, milliseconds to seconds
Spin rule	Spin-allowed	Spin-forbidden, so slow
Observation	Stops quickly when source is removed	May continue after source is removed

Previous-Year Style Questions

1. Explain selection rules for rotational and vibrational spectroscopy.
2. Derive the expression for vibrational frequency of a diatomic molecule.
3. Explain fluorescence with Jablonski diagram and give medical applications.
4. State Beer-Lambert law and solve a concentration-based numerical.
5. Derive Bragg law and explain its use in crystal structure determination.

2.10 Short Notes on Surface Characterisation

Surface characterisation is used to study the outermost layers of solids, thin films, catalysts, sensors and biomaterials. These methods are important in engineering because surface properties often decide corrosion resistance, adhesion, catalytic activity and device performance.

Technique	Information obtained	Engineering use
SEM	Surface morphology and particle shape	Fracture surface, coating quality, microstructure
AFM	Nanometre-scale surface topography	Thin films, MEMS surfaces, biomaterials
XPS	Surface elemental composition and oxidation state	Corrosion films, catalysts, semiconductor surfaces
XRD	Crystal structure and interplanar spacing	Phase identification and crystallinity

Exam Answer: Why Surface Analysis Matters

Bulk composition may remain unchanged while the surface oxidizes, adsorbs impurities or forms a thin passive layer. Therefore surface techniques are needed to understand corrosion, catalysis, coating failure and sensor response.

Quick Revision — Module 2

Spectroscopic Techniques and Applications | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\Delta E = h\nu = \frac{hc}{\lambda}$	Photon energy absorbed or emitted in spectroscopy.
$\Delta J = \pm 1$	Rotational selection rule.
$\Delta v = \pm 1$	Fundamental vibrational transition rule.
$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$	Diatomic vibrational frequency.
$\mu = \frac{m_1 m_2}{m_1 + m_2}$	Reduced mass of a diatomic molecule.
$n\lambda = 2d \sin \theta$	Bragg equation for diffraction.

One-Line Definitions

- **Spectroscopy:** Study of interaction of radiation with matter.
- **Selection rule:** Condition deciding whether a transition is allowed.
- **Chromophore:** Group responsible for UV-visible absorption.
- **Fluorescence:** Fast emission after electronic excitation.
- **IR active vibration:** Vibration causing a change in dipole moment.
- **Chemical shift:** NMR position relative to a reference.
- **Diffraction:** Constructive interference of waves scattered by ordered planes.

Frequently Asked / Viva Questions

Q1. What is spectroscopy?

It is the study of absorption, emission, or scattering of radiation by matter.

Q2. Write the relation between energy and wavelength.

$$\Delta E = hc/\lambda.$$

Q3. What is a selection rule?

It is a rule that decides whether a spectroscopic transition is allowed.

Q4. Why are rotational spectra shown by polar molecules?

Rotation must produce an oscillating dipole moment.

Q5. What is the condition for IR activity?

The vibration must change the dipole moment of the molecule.

Frequently Asked / Viva Questions (contd.)

Q6. What is fluorescence?

It is emission of light from an excited singlet state to the ground state.

Q7. Give one medical use of fluorescence.

Fluorescent probes are used in bio-imaging and disease diagnostics.

Q8. What is NMR based on?

Resonance absorption by nuclei with non-zero spin in a magnetic field.

Q9. How is MRI related to NMR?

MRI is an imaging technique based on NMR signals, mainly from hydrogen nuclei.

Q10. State Bragg law.

$$n\lambda = 2d \sin \theta.$$

Summary Reference Table

Method	Radiation / signal	Main application
UV-visible	Electronic transitions	Conjugation and concentration
Fluorescence	Emitted visible light	Medical imaging and trace analysis
IR	Vibrational transitions	Functional group identification
NMR/MRI	Nuclear spin resonance	Structure and medical imaging
XRD	Diffacted X-rays	Crystal structure

Module 3

Intermolecular Forces and Potential Energy Surfaces

1. Intermolecular Forces and Potential Energy Surfaces

Intermolecular forces are attractive or repulsive forces between molecules or ions. They are weaker than covalent bonds but strongly affect boiling point, solubility, viscosity, and real-gas behaviour.

1.1 Ionic, Dipole and Van der Waals Interactions

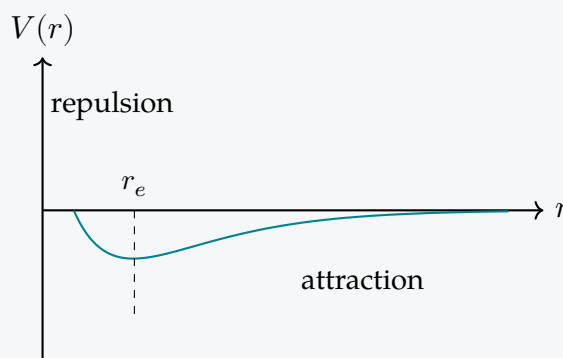
Ionic interactions are strongest among common non-covalent interactions. Dipole-dipole forces occur between polar molecules. London dispersion forces arise from temporary induced dipoles and exist in all molecules.

Interaction	Origin	Example
Ion-ion	Attraction between full charges	Na^+ and Cl^-
Ion-dipole	Ion attracts polar molecule	Hydration of Na^+
Dipole-dipole	Permanent dipoles align	HCl molecules
London dispersion	Instantaneous induced dipoles	Noble gases, hydrocarbons
Hydrogen bonding	H attached to F, O, or N interacts with lone pair	Water, alcohols

1.2 Potential Energy Curve

The potential energy between two particles depends on separation distance. At equilibrium distance r_e , attractive and repulsive forces balance and potential energy is minimum.

Potential Energy Curve for Intermolecular Interaction



1.3 Equations of State for Real Gases

Ideal gas equation assumes no intermolecular force and zero molecular volume. Real gases deviate at high pressure and low temperature.

Van der Waals Equation

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

where a corrects attractive forces and b corrects finite molecular volume.

1.4 Critical Phenomena

The critical temperature T_c is the temperature above which a gas cannot be liquefied by pressure alone. At the critical point, liquid and vapour become indistinguishable.

Critical Constants for a Van der Waals Gas

$$V_c = 3nb, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

Exam Points

- Compare intermolecular forces by strength and origin.
- Draw the potential energy curve and mark r_e and minimum energy.
- For real gases, explain physical meaning of constants a and b .
- Define critical temperature, pressure, and volume clearly.

2. Detailed Notes on Forces, Real Gases and Critical Behaviour

2.1 Origin and Strength of Intermolecular Forces

Intermolecular force strength depends on charge, polarity, polarizability, and molecular shape. Ionic interactions are strong because full charges are involved. Dispersion forces are weak individually but become important for large molecules with many electrons.

Force	Approximate strength trend	Property affected
Ion-ion	Very strong	Melting point of ionic solids
Ion-dipole	Strong	Hydration and solubility of salts
Hydrogen bond	Moderate to strong	Boiling point of water and alcohols
Dipole-dipole	Moderate	Boiling point of polar molecules
London dispersion	Weak to moderate	Liquefaction of non-polar gases

2.2 Potential Energy and Force Relation

The force between two particles is related to the potential energy curve by:

$$F = -\frac{dV}{dr}$$

At the equilibrium separation r_e , potential energy is minimum, so:

$$\left(\frac{dV}{dr}\right)_{r=r_e} = 0, \quad F = 0$$

For $r < r_e$, repulsive force dominates. For $r > r_e$, attractive force dominates.

Exam Interpretation of Potential Energy Curve

If two molecules are very far apart, $V(r) \approx 0$. As they approach, attraction lowers the potential energy. At r_e , the most stable separation is reached. If they come closer than r_e , electron-cloud repulsion sharply increases energy.

2.3 Lennard-Jones Potential

A common model for non-bonded interaction is:

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right]$$

The r^{-12} term represents short-range repulsion and the r^{-6} term represents London attraction. ϵ indicates well depth, i.e. strength of attraction.

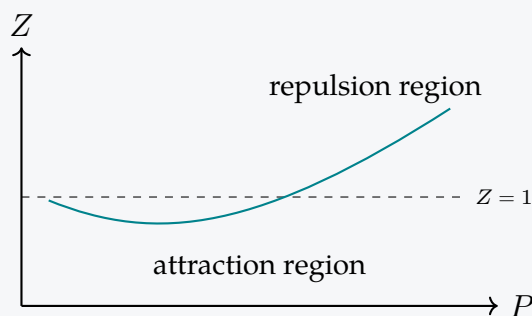
Term	Physical meaning
$(\sigma/r)^{12}$	Repulsion due to electron-cloud overlap
$-(\sigma/r)^6$	Attractive dispersion interaction
ϵ	Depth of potential well; higher value means stronger attraction
r_e	Separation at minimum potential energy

2.4 Compressibility Factor and Deviation from Ideality

Deviation from ideal gas behaviour is often measured using compressibility factor:

$$Z = \frac{PV}{nRT}$$

For an ideal gas, $Z = 1$. For a real gas, $Z < 1$ indicates attraction-dominated behaviour and $Z > 1$ indicates repulsion or finite-volume dominated behaviour.

Compressibility Factor: Qualitative Behaviour**2.5 Critical Constants from Van der Waals Equation**

For one mole:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

At the critical point, the isotherm has a point of inflection:

$$\left(\frac{\partial P}{\partial V_m}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial V_m^2}\right)_T = 0$$

Solving these conditions gives:

$$V_c = 3b, \quad P_c = \frac{a}{27b^2}, \quad T_c = \frac{8a}{27Rb}$$

These relations are often asked as short derivation or formula-based questions.

Solved Example: Critical Volume

If van der Waals constant $b = 0.040 \text{ L mol}^{-1}$, then:

$$V_c = 3b = 3(0.040) = 0.120 \text{ L mol}^{-1}$$

How to Score in This Module

- Write force origin, example, and property affected for each intermolecular force.
- In potential energy questions, always mark attraction, repulsion, r_e , and energy minimum.
- For real gases, explain that a corrects pressure and b corrects volume.
- Use $Z = PV/nRT$ to discuss positive and negative deviation.

2.6 Hydrogen Bonding in Detail

Hydrogen bonding occurs when hydrogen is covalently bonded to a highly electronegative atom such as F, O, or N and interacts with a lone pair on another electronegative atom.

Type	Meaning	Example
Intermolecular hydrogen bonding	Hydrogen bond between different molecules	Water, alcohols, HF
Intramolecular hydrogen bonding	Hydrogen bond within the same molecule	o-nitrophenol type structures
Strong hydrogen bonding	Shorter and more directional interaction	HF, water network
Weak hydrogen bonding	Longer and less directional interaction	C–H based weak interactions

Exam Explanation: Boiling Point of Water

Water has a much higher boiling point than H_2S because oxygen is highly electronegative and forms extensive intermolecular hydrogen bonding. More heat is required to break this associated network before molecules escape into vapour phase.

2.7 Corresponding States and Reduced Variables

The law of corresponding states states that gases behave similarly when compared at the same reduced temperature and reduced pressure.

Reduced Variables

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}, \quad V_r = \frac{V}{V_c}$$

where T_c , P_c , and V_c are critical constants.

2.8 Boyle Temperature

At Boyle temperature, a real gas behaves nearly ideally over a considerable pressure range because attractive and repulsive deviations approximately cancel.

$$Z \approx 1$$

For a van der Waals gas, the Boyle temperature is approximately:

$$T_B = \frac{a}{Rb}$$

Solved Example: Boyle Temperature

If $a = 1.36 \text{ L}^2 \text{ atm mol}^{-2}$ and $b = 0.0318 \text{ L mol}^{-1}$:

$$T_B = \frac{a}{Rb} = \frac{1.36}{(0.0821)(0.0318)}$$

$$T_B \approx 521 \text{ K}$$

2.9 Joule-Thomson Effect

When a real gas expands through a porous plug from high pressure to low pressure without external work and without heat exchange, its temperature may change. This is called

Joule-Thomson effect.

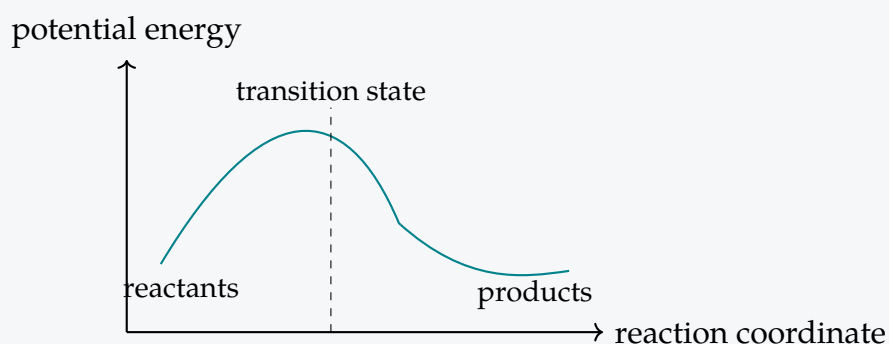
Inversion temperature: The temperature at which Joule-Thomson coefficient becomes zero. Below it, gas cools on expansion; above it, gas warms on expansion.

Condition	Joule-Thomson coefficient	Observation
$T < T_i$	Positive	Gas cools on expansion
$T = T_i$	Zero	No temperature change
$T > T_i$	Negative	Gas warms on expansion

2.10 Potential Energy Surface in Chemical Reactions

A potential energy surface shows how molecular energy changes with geometry. Reactants, products, intermediates, and transition states are represented as regions on the surface.

Reaction Path on a Potential Energy Surface



Previous-Year Style Questions

1. Compare ion-dipole, dipole-dipole, hydrogen bonding and dispersion forces.
2. Explain the potential energy curve for two interacting molecules.
3. Derive or state critical constants of a van der Waals gas.
4. Explain compressibility factor and deviation from ideal behaviour.
5. Write short notes on Joule-Thomson effect and inversion temperature.

2.11 Applications of Intermolecular Forces in Engineering Chemistry

Intermolecular forces are not only theoretical; they explain important material and chemical behaviour.

Application	Force involved	Importance
Solubility of salts in water	Ion-dipole force	Hydration stabilizes ions in solution
Polymer strength	Dipole interaction and hydrogen bonding	Controls toughness, flexibility and melting range
Adsorption on catalyst	van der Waals and polar interactions	Reactants attach to active surface sites
Boiling point of liquids	Hydrogen bonding and dispersion	Stronger forces increase boiling point
Protein structure	Hydrogen bonding and hydrophobic interaction	Maintains secondary and tertiary structure

Exam Answer: Like Dissolves Like

Polar solvents dissolve ionic and polar solutes because ion-dipole or dipole-dipole interactions compensate for the energy needed to separate solute particles. Non-polar solutes dissolve better in non-polar solvents because dispersion forces are similar in both.

Quick Revision — Module 3

Intermolecular Forces and Potential Energy Surfaces | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$F = -\frac{dV}{dr}$	Force is negative gradient of potential energy.
$PV = nRT$	Ideal gas equation.
$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$	Van der Waals real-gas equation.
$V_c = 3nb$	Critical volume for van der Waals gas.
$P_c = \frac{a}{27b^2}$	Critical pressure for one mole form.
$T_c = \frac{8a}{27Rb}$	Critical temperature for one mole form.

One-Line Definitions

- **Intermolecular force:** Force acting between molecules or ions.
- **Dipole moment:** Product of charge and separation in a polar molecule.
- **Hydrogen bond:** Strong dipole interaction involving H bonded to F, O, or N.
- **Potential energy surface:** Energy variation with molecular geometry.
- **Real gas:** Gas that deviates from ideal behaviour due to volume and attraction.
- **Critical temperature:** Highest temperature at which gas can be liquefied by pressure.

Frequently Asked / Viva Questions

Q1. What are intermolecular forces?

Forces of attraction or repulsion acting between separate molecules or ions.

Q2. Arrange common forces in increasing strength.

London dispersion < dipole-dipole < hydrogen bonding < ion-dipole < ion-ion.

Q3. Why does boiling point increase with stronger intermolecular force?

More energy is needed to separate molecules into vapour.

Q4. What is equilibrium separation in a potential energy curve?

It is the distance where potential energy is minimum and net force is zero.

Q5. What causes repulsion at very small distance?

Overlap of electron clouds and nuclear repulsion.

Frequently Asked / Viva Questions (contd.)

Q6. Why do real gases deviate from ideal behaviour?

Molecules have finite volume and attractive forces.

Q7. What is the role of a in van der Waals equation?

It corrects pressure for intermolecular attraction.

Q8. What is the role of b in van der Waals equation?

It corrects volume for finite molecular size.

Q9. Define critical point.

The point at which liquid and vapour phases become indistinguishable.

Q10. What is critical temperature?

The highest temperature above which gas cannot be liquefied by pressure alone.

Summary Reference Table

Concept	Meaning	Must write in exam
Intermolecular forces	Non-covalent forces between species	Strength order and examples
Potential curve	Energy versus separation	Minimum at r_e
Real gas	Non-ideal gas	Van der Waals correction terms
Critical point	End point of liquid-vapour boundary	T_c, P_c, V_c definitions

Module 4

Free Energy and Chemical Equilibria

1. Free Energy and Chemical Equilibria

Gibbs free energy is the thermodynamic function that predicts spontaneity at constant temperature and pressure. A process is spontaneous when $\Delta G < 0$.

1.1 First and Second Laws of Thermodynamics

The first law states conservation of energy. The second law states that total entropy of the universe increases for a spontaneous process.

Thermodynamic Functions

$$\Delta U = q + w, \quad \Delta H = \Delta U + \Delta(PV)$$

$$\Delta G = \Delta H - T\Delta S$$

At constant temperature and pressure, $\Delta G < 0$ means spontaneous, $\Delta G = 0$ means equilibrium, and $\Delta G > 0$ means non-spontaneous.

1.2 Entropy and Free Energy Estimation

Entropy measures dispersal of energy. For a reversible process,

$$\Delta S = \int \frac{\delta q_{\text{rev}}}{T}$$

Free energy change combines enthalpy and entropy and is useful because it directly predicts useful work and equilibrium direction.

1.3 Free Energy and Equilibrium Constant

For a reaction at non-standard condition,

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium, $\Delta G = 0$ and $Q = K$.

Relation Between Free Energy and Equilibrium Constant

$$\Delta G^\circ = -RT \ln K = -2.303RT \log K$$

1.4 Cell Potential and Nernst Equation

Electrochemical cell work is linked with free energy.

Electrochemical Free Energy

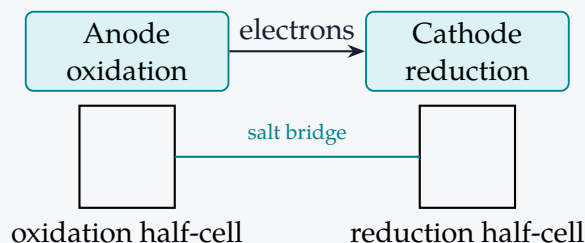
$$\Delta G = -nFE, \quad \Delta G^\circ = -nFE^\circ$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

Galvanic Cell and Electron Flow



1.5 Acid-Base, Redox and Solubility Equilibria

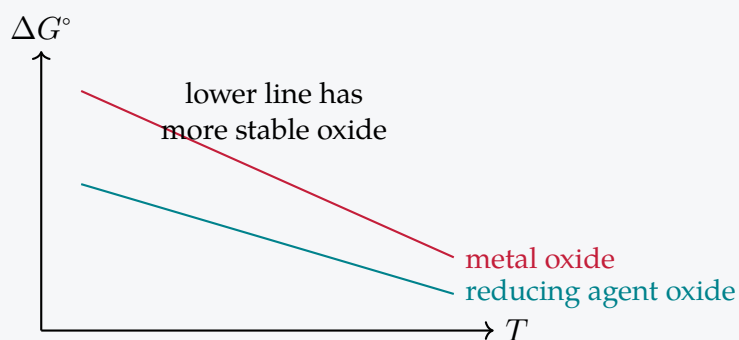
The same free-energy idea applies to acid-base ionization, redox reactions, and precipitation.

Equilibrium	Constant	Use
Acid-base	K_a, K_b	pH and strength of acid/base
Redox	E°, K	Feasibility of electron transfer
Solubility	K_{sp}	Precipitation and ionic product

1.6 Water Chemistry, Corrosion and Ellingham Diagram

Water chemistry includes hardness, alkalinity, pH, and dissolved oxygen. Corrosion is an electrochemical oxidation of metal. Ellingham diagrams plot ΔG° versus temperature and help choose reducing agents in metallurgy.

Ellingham Diagram Concept



Exam Points

- Always state sign of ΔG for spontaneity.
- For equilibrium questions, start from $\Delta G = \Delta G^\circ + RT \ln Q$.
- For electrochemistry, connect ΔG , E , and Nernst equation.
- For Ellingham diagrams, lower line means more stable oxide.

2. Derivations, Numericals and Engineering Applications

2.1 Deriving the Equilibrium Relation

For any reaction:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium, the reaction has no net driving force, so:

$$\Delta G = 0, \quad Q = K$$

Therefore:

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

Using $\ln K = 2.303 \log K$:

$$\Delta G^\circ = -2.303RT \log K$$

Solved Example: Equilibrium Constant from Free Energy

At 298 K, suppose $\Delta G^\circ = -11.4 \text{ kJ mol}^{-1}$.

$$\log K = \frac{-\Delta G^\circ}{2.303RT}$$

$$\log K = \frac{11400}{2.303(8.314)(298)} \approx 2.00$$

$$K \approx 10^2 = 100$$

Since $K > 1$, products are favoured.

2.2 Entropy Change for Common Processes

Entropy increases when randomness, volume, or number of gas molecules increases.

Process	Entropy change	Reason
Melting of solid	Positive	Liquid has more molecular freedom
Vaporization	Strongly positive	Gas has much greater randomness
Compression of gas	Negative	Accessible volume decreases
Dissolution of salt	Usually positive	Ions disperse in solution
Crystallization	Negative	Ordered solid forms

2.3 Nernst Equation Derivation from Free Energy

Electrical work from a cell is:

$$\Delta G = -nFE$$

At standard condition:

$$\Delta G^\circ = -nFE^\circ$$

Substitute both in:

$$\begin{aligned}\Delta G &= \Delta G^\circ + RT \ln Q \\ -nFE &= -nFE^\circ + RT \ln Q\end{aligned}$$

Dividing by $-nF$:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

Solved Example: Cell Potential

For a cell with $E^\circ = 1.10$ V, $n = 2$, and $Q = 10$:

$$\begin{aligned}E &= 1.10 - \frac{0.0591}{2} \log 10 \\ E &= 1.10 - 0.02955 = 1.07045 \text{ V}\end{aligned}$$

2.4 Water Hardness and Boiler Problems

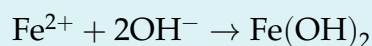
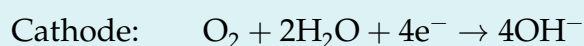
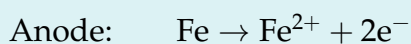
Hard water contains dissolved calcium and magnesium salts. Temporary hardness is due to bicarbonates; permanent hardness is due to chlorides and sulphates.

Problem	Cause	Effect in engineering systems
Scale formation	Deposition of salts like CaCO_3	Poor heat transfer and tube overheating
Sludge formation	Loose precipitates in boiler water	Choking and reduced efficiency
Priming	Carryover of water droplets with steam	Wet steam and turbine damage
Foaming	Stable bubbles due to impurities	Irregular boiler operation
Caustic embrittlement	Concentrated alkali in cracks	Boiler metal failure

2.5 Electrochemical Corrosion Mechanism

In iron corrosion, anodic and cathodic regions form on the metal surface.

Rusting Half-Reactions



Further oxidation gives hydrated ferric oxide, commonly called rust.

2.6 Ellingham Diagram: Reduction Rule

In metallurgy, a metal oxide can be reduced by an element whose oxide line lies below it in the Ellingham diagram. This means the reducing agent forms a more stable oxide and makes the overall ΔG° negative.

Answer-Writing Pattern for Ellingham Problems

1. Identify the oxide to be reduced.
2. Locate its ΔG° line at the given temperature.
3. Find a reducing agent whose oxide line is lower.
4. State that the net free energy change becomes negative.

2.7 Temperature Dependence of Spontaneity

The sign of $\Delta G = \Delta H - T\Delta S$ depends on both enthalpy and entropy. This table is a common 5-mark answer.

Case	Sign condition	Spontaneity
Exothermic, entropy increases	$\Delta H < 0, \Delta S > 0$	Spontaneous at all temperatures
Endothermic, entropy decreases	$\Delta H > 0, \Delta S < 0$	Non-spontaneous at all temperatures
Exothermic, entropy decreases	$\Delta H < 0, \Delta S < 0$	Spontaneous at low temperature
Endothermic, entropy increases	$\Delta H > 0, \Delta S > 0$	Spontaneous at high temperature

Solved Example: Temperature of Spontaneity

For a reaction, $\Delta H = 40 \text{ kJ mol}^{-1}$ and $\Delta S = 100 \text{ J mol}^{-1} \text{ K}^{-1}$. Convert entropy:

$$\Delta S = 0.100 \text{ kJ mol}^{-1} \text{ K}^{-1}$$

At equilibrium, $\Delta G = 0$:

$$T = \frac{\Delta H}{\Delta S} = \frac{40}{0.100} = 400 \text{ K}$$

Since both ΔH and ΔS are positive, the reaction is spontaneous above 400 K.

2.8 pH, Acid Dissociation and Buffer Idea

For a weak acid:



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

For a buffer made of weak acid and its salt:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Solved Example: Buffer pH

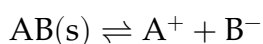
If $\text{p}K_a = 4.75$, $[\text{salt}] = 0.20 \text{ M}$ and $[\text{acid}] = 0.10 \text{ M}$:

$$\text{pH} = 4.75 + \log \frac{0.20}{0.10}$$

$$\text{pH} = 4.75 + \log 2 = 4.75 + 0.301 = 5.051$$

2.9 Solubility Product and Common Ion Effect

For a sparingly soluble salt AB:



$$K_{sp} = [A^+][B^-]$$

Precipitation occurs when ionic product exceeds K_{sp} .

Condition	Meaning	Observation
Ionic product $< K_{sp}$	Unsaturated solution	No precipitation
Ionic product $= K_{sp}$	Saturated solution	Equilibrium
Ionic product $> K_{sp}$	Supersaturated with respect to salt	Precipitation occurs

2.10 Corrosion Prevention Methods

Corrosion can be reduced by blocking contact with environment or by making the metal cathodic.

Method	Principle	Example
Painting / coating	Barrier between metal and environment	Paint, oil, polymer coating
Galvanization	Zinc corrodes sacrificially before iron	Zn-coated iron sheets
Cathodic protection	Metal to be protected is made cathode	Pipelines, ship hulls
Alloying	Forms corrosion-resistant surface film	Stainless steel
Inhibitors	Reduce anodic or cathodic reaction rate	Boiler-water additives

Previous-Year Style Questions

1. Derive $\Delta G^\circ = -RT \ln K$ and explain its significance.
2. Derive Nernst equation and solve a cell-potential numerical.
3. Explain water hardness, scale formation and boiler troubles.
4. Describe electrochemical corrosion of iron and prevention methods.
5. Explain Ellingham diagram and its use in metallurgy.

Quick Revision — Module 4

Free Energy and Chemical Equilibria | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\Delta U = q + w$	First law of thermodynamics.
$\Delta G = \Delta H - T\Delta S$	Gibbs free energy relation.
$\Delta G = \Delta G^\circ + RT \ln Q$	Free energy at non-standard condition.
$\Delta G^\circ = -RT \ln K$	Relation between free energy and equilibrium constant.
$\Delta G = -nFE$	Cell free energy and cell potential.
$E = E^\circ - \frac{RT}{nF} \ln Q$	Nernst equation.
$K_{sp} = [M^{m+}]^n [X^{n-}]^m$	Solubility product expression.

One-Line Definitions

- **Entropy:** Measure of energy dispersal or randomness.
- **Gibbs free energy:** Energy available for useful work at constant temperature and pressure.
- **Equilibrium constant:** Ratio of product and reactant activities at equilibrium.
- **Nernst equation:** Relation between electrode potential and concentration.
- **Corrosion:** Electrochemical deterioration of a metal.
- **Ellingham diagram:** Plot of ΔG° for oxide formation versus temperature.

Frequently Asked / Viva Questions

Q1. State the first law of thermodynamics.

Energy cannot be created or destroyed; $\Delta U = q + w$.

Q2. What is the criterion of spontaneity using Gibbs free energy?

$\Delta G < 0$ spontaneous, $\Delta G = 0$ equilibrium, $\Delta G > 0$ non-spontaneous.

Q3. Write Gibbs free energy equation.

$\Delta G = \Delta H - T\Delta S$.

Q4. Relate ΔG° and equilibrium constant.

$\Delta G^\circ = -RT \ln K$.

Q5. How is cell emf related to free energy?

$\Delta G = -nFE$.

Frequently Asked / Viva Questions (contd.)

Q6. State the Nernst equation at 298 K.

$$E = E^{\circ} - (0.0591/n) \log Q.$$

Q7. What is solubility product?

It is the product of ion concentrations in a saturated solution, each raised to stoichiometric power.

Q8. Why does corrosion occur electrochemically?

Metal oxidation and reduction of oxygen or hydrogen occur at different surface sites.

Q9. What does an Ellingham diagram show?

It shows temperature dependence of standard free energy of oxide formation.

Q10. How is a reducing agent selected from Ellingham diagram?

A reducing agent whose oxide line lies below the metal oxide line can reduce that oxide.

Summary Reference Table

Area	Key relation	Exam focus
Thermodynamics	$\Delta G = \Delta H - T\Delta S$	Spontaneity and equilibrium
Chemical equilibrium	$\Delta G^{\circ} = -RT \ln K$	Direction and extent of reaction
Electrochemistry	$\Delta G = -nFE$	Cell potential and Nernst equation
Metallurgy	Ellingham diagram	Reduction feasibility

Module 5

Periodic Properties

1. Periodic Properties

Periodic properties are atomic and chemical properties that vary systematically with atomic number due to effective nuclear charge, shielding, and electronic configuration.

1.1 Effective Nuclear Charge and Shielding

Outer electrons do not experience the full nuclear charge because inner electrons shield them. Effective nuclear charge is:

Effective Nuclear Charge

$$Z_{\text{eff}} = Z - \sigma$$

where Z is nuclear charge and σ is shielding constant.

1.2 Penetration and Orbital Energy

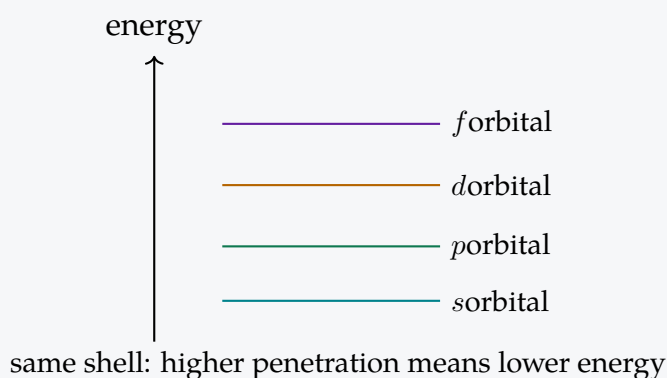
For the same principal shell, penetration order is:

$$s > p > d > f$$

More penetrating orbitals experience higher Z_{eff} and lower energy. Therefore in multi-electron atoms:

$$E(ns) < E(np) < E(nd) < E(nf)$$

Penetration and Energy Order



1.3 Atomic Size, Ionization Energy, Electron Affinity and Electronegativity

Across a period, Z_{eff} increases, so atomic size decreases and ionization energy generally increases. Down a group, new shells are added, so atomic size increases and ionization energy decreases.

Property	Across period	Down group
Atomic radius	Decreases	Increases
Ionization energy	Generally increases	Decreases
Electron affinity	Becomes more negative generally	Less negative generally
Electronegativity	Increases	Decreases
Metallic character	Decreases	Increases

1.4 Polarizability, Oxidation States and Coordination

Polarizability increases with size and diffuse electron cloud. Oxidation state depends on valence electron configuration. Coordination number and geometry depend on metal ion size, ligand size, and electron count.

Coordination number	Common geometry	Example type
2	Linear	$[\text{Ag}(\text{NH}_3)_2]^+$
4	Tetrahedral or square planar	$[\text{Ni}(\text{CN})_4]^{2-}$
6	Octahedral	$[\text{Fe}(\text{CN})_6]^{3-}$

1.5 Hard and Soft Acids and Bases

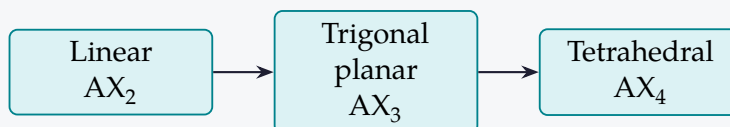
HSAB principle says hard acids prefer hard bases and soft acids prefer soft bases.

Hard species are small, highly charged, and weakly polarizable. **Soft species** are larger, lower charged, and highly polarizable.

1.6 Molecular Geometries

VSEPR theory predicts geometry from electron-pair repulsion around the central atom.

Common VSEPR Shapes



Exam Points

- Explain periodic trends using Z_{eff} , shielding, and shell number.
- For orbital energy, mention penetration order $s > p > d > f$.

Frequently Asked / Viva Questions (contd.)

- HSAB questions need definition plus examples.
- Geometry questions should mention electron pairs before shape.

2. Detailed Periodic Trend Analysis

2.1 Shielding Order and Slater-Type Reasoning

Shielding is not equal for all orbitals. Inner-shell electrons shield strongly, while electrons in the same shell shield only partially. Poor shielding by *d* and *f* electrons explains many irregular periodic trends.

Electron type	Shielding ability	Consequence
<i>s</i> electron	Strong penetration and good shielding	Feels high Z_{eff}
<i>p</i> electron	Moderate shielding	Higher energy than same-shell <i>s</i>
<i>d</i> electron	Poor shielding	Causes transition-series contraction
<i>f</i> electron	Very poor shielding	Causes lanthanide contraction

2.2 Successive Ionization Energies

Successive ionization energies always increase because after removal of one electron, the remaining electrons are held more strongly by the positive ion.

$$\text{IE}_1 < \text{IE}_2 < \text{IE}_3 < \dots$$

A sudden jump in ionization energy indicates removal of an electron from an inner shell.

Solved Example: Identifying Valence Electrons

If successive ionization energies show the pattern:

$$580, \quad 1820, \quad 2750, \quad 11600 \text{ kJ mol}^{-1}$$

the large jump occurs after the third electron. Therefore the atom has **three valence electrons**.

2.3 Diagonal Relationship

Some second-period elements resemble third-period elements diagonally placed in the periodic table. Examples include Li–Mg, Be–Al, and B–Si. This occurs because size and charge density become comparable.

Pair	Common similarities	Reason
Li and Mg	Form nitrides; carbonates decompose on heating	Similar charge density
Be and Al	Amphoteric oxides and covalent halides	Small size and high polarizing power
B and Si	Covalent compounds and acidic oxides	Similar electronegativity trend

2.4 Oxidation States and Inert Pair Effect

For heavier *p*-block elements, the outer ns^2 electron pair becomes less available for bonding. This is called inert pair effect. It stabilizes lower oxidation states.

Inert pair effect: Reluctance of the outermost ns^2 electron pair of heavier elements to participate in bonding.

Group	Higher oxidation state	Lower oxidation state stabilized down group
Group 13	+3	+1 in Tl
Group 14	+4	+2 in Pb
Group 15	+5	+3 in Bi

2.5 HSAB Applications

HSAB theory predicts stability of complexes and salts. Hard-hard and soft-soft combinations are more stable than hard-soft combinations.

Acid / base type	Examples	Preferred partner
Hard acids	H^+ , Li^+ , Al^{3+} , Fe^{3+}	Hard bases
Soft acids	Ag^+ , Hg^{2+} , Pt^{2+}	Soft bases
Hard bases	F^- , OH^- , H_2O , NH_3	Hard acids
Soft bases	I^- , S^{2-} , CO , PR_3	Soft acids

Solved Example: HSAB Prediction

Ag^+ is a soft acid. Between F^- and I^- , iodide is softer. Therefore AgI is more stable and less soluble than expected from simple charge arguments.

2.6 VSEPR Shapes and Lone Pair Effect

Lone pairs repel more strongly than bond pairs. The repulsion order is:

$$\text{lone pair-lone pair} > \text{lone pair-bond pair} > \text{bond pair-bond pair}$$

Molecule type	Shape	Example
AX_2	Linear	$BeCl_2$, CO_2
AX_3	Trigonal planar	BF_3
AX_3E	Trigonal pyramidal	NH_3
AX_2E_2	Bent	H_2O
AX_4	Tetrahedral	CH_4

Periodic Properties: 10-Mark Answer Framework

1. Define the property.
2. State the trend across period and down group.
3. Explain trend using Z_{eff} , shielding, and shell size.
4. Mention exceptions if relevant.
5. Add one example or application such as HSAB or VSEPR shape.

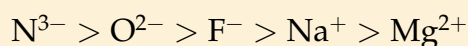
2.7 Atomic and Ionic Radius: Detailed Trend

Atomic radius decreases across a period because effective nuclear charge increases while electrons enter the same shell. Down a group, radius increases because a new shell is added.

Species compared	Trend	Reason
Cations vs atoms	Cations are smaller	Electron loss reduces repulsion and may remove outer shell
Anions vs atoms	Anions are larger	Extra electron increases electron-electron repulsion
Isoelectronic species	Radius decreases with increasing nuclear charge	Same electrons pulled by larger Z

Solved Example: Isoelectronic Radius Order

N^{3-} , O^{2-} , F^- , Na^+ and Mg^{2+} are isoelectronic with 10 electrons.
Higher nuclear charge pulls the same electron cloud more strongly:



2.8 Electron Affinity and Exceptions

Electron affinity generally becomes more negative across a period, but exceptions occur due to stable half-filled or filled configurations and small size.

Exception	Observation	Reason
N less negative than C/O	N resists electron gain	Stable half-filled $2p^3$ configuration
F less negative than Cl	Cl has more negative electron affinity	Very small F atom has high electron-electron repulsion
Noble gases	Positive or very low tendency	Stable closed shell

2.9 Electronegativity Scales

Electronegativity is not directly measured; it is assigned using scales. The Pauling scale is most common in basic chemistry.

Scale	Basis	Use
Pauling	Bond energy differences	Most common qualitative scale
Mulliken	Average of ionization energy and electron affinity	Connects to atomic properties
Allred-Rochow	Electrostatic attraction by nucleus	Uses effective nuclear charge and radius

2.10 Polarization and Fajan's Rules

Fajan's rules explain covalent character in ionic compounds. A small highly charged cation polarizes a large anion strongly, increasing covalent character.

Fajan's Rules

- Smaller cation has greater polarizing power.
- Higher cation charge has greater polarizing power.
- Larger anion is more easily polarized.
- Cations with pseudo-noble gas configuration polarize more strongly.

Solved Example: Covalent Character

AlCl_3 has more covalent character than NaCl because Al^{3+} is smaller and more highly charged than Na^+ , so it polarizes Cl^- more strongly.

2.11 Coordination Number and Geometry in Complexes

Coordination number is the number of ligand donor atoms directly attached to the metal ion. Geometry depends on coordination number and electronic arrangement.

Coordination number	Geometry	Typical comment
2	Linear	Common for Ag^+ and Cu^+ complexes
4	Tetrahedral	Common when ligands are bulky or weak-field
4	Square planar	Common for d^8 metal ions like Pt^{2+}
6	Octahedral	Most common geometry for transition-metal complexes

Previous-Year Style Questions

1. Explain effective nuclear charge and shielding with periodic trends.
2. Discuss trends in atomic radius, ionization energy and electronegativity.
3. Explain penetration of s , p , d , f orbitals and its effect on energy.
4. State HSAB principle and predict stable acid-base combinations.
5. Explain VSEPR shapes and coordination geometries with examples.

2.12 Common Exceptions and Exam Traps

Periodic trends are general rules, but exam questions often test exceptions. Mentioning the reason behind exceptions improves answer quality.

Trend exception	Observation	Reason
Be and B ionization energy	$\text{IE}(\text{Be}) > \text{IE}(\text{B})$	Electron removed from B enters higher-energy $2p$ orbital
N and O ionization energy	$\text{IE}(\text{N}) > \text{IE}(\text{O})$	N has stable half-filled $2p^3$ arrangement
Ga smaller than Al anomaly	Size does not increase as expected	Poor shielding by filled $3d$ electrons
Lanthanide contraction	Similar radii of $4d$ and $5d$ elements	Poor shielding by $4f$ electrons

Exam Answer: Why $\text{IE}(\text{N}) > \text{IE}(\text{O})$?

Nitrogen has electronic configuration $1s^2 2s^2 2p^3$, where the three p electrons are half-filled and relatively stable. Oxygen has $1s^2 2s^2 2p^4$, so one p orbital contains paired electrons. Removing one electron from oxygen reduces electron-electron repulsion, so oxygen has lower first ionization energy than nitrogen.

Quick Revision — Module 5

Periodic Properties | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$Z_{\text{eff}} = Z - \sigma$	Effective nuclear charge felt by valence electron.
$s > p > d > f$	Penetration order within a shell.
$E(ns) < E(np) < E(nd) < E(nf)$	Orbital energy order for same principal shell in multi-electron atoms.
$IE_1 < IE_2 < IE_3$	Successive ionization energies increase.
$\mu = \alpha E$	Induced dipole moment relation; α is polarizability.

One-Line Definitions

- **Effective nuclear charge:** Net positive charge experienced by an electron.
- **Shielding:** Reduction of nuclear attraction due to inner electrons.
- **Penetration:** Ability of an orbital to approach the nucleus.
- **Ionization energy:** Energy required to remove an electron from gaseous atom.
- **Electron affinity:** Energy change when an electron is added to gaseous atom.
- **Electronegativity:** Tendency of atom to attract bonded electron pair.
- **HSAB principle:** Hard acids bind hard bases and soft acids bind soft bases.

Frequently Asked / Viva Questions

Q1. Define effective nuclear charge.

It is the net nuclear charge felt by an electron after shielding.

Q2. Why does atomic radius decrease across a period?

Z_{eff} increases and pulls electrons closer.

Q3. Why does ionization energy decrease down a group?

Atomic size and shielding increase, so valence electrons are removed more easily.

Q4. Write the penetration order.

$s > p > d > f$.

Q5. Why is an s orbital lower in energy than a p orbital of same shell?

The s orbital penetrates closer to the nucleus and feels greater Z_{eff} .

Q6. What is polarizability?

Frequently Asked / Viva Questions (contd.)

Ease with which an electron cloud is distorted by an electric field.

Q7. What is HSAB principle?

Hard acids prefer hard bases and soft acids prefer soft bases.

Q8. Give examples of hard and soft bases.

Hard bases: F^- , OH^- ; soft bases: I^- , S^{2-} .

Q9. What decides coordination geometry?

Coordination number, ligand field, metal ion size, and electron configuration.

Q10. What is VSEPR theory used for?

It predicts molecular geometry from repulsion among electron pairs.

Summary Reference Table

Trend	Reason	Result
Across period	Z_{eff} increases	Size decreases, IE increases
Down group	Shell number increases	Size increases, IE decreases
Penetration	s enters closest to nucleus	s orbital has lower energy
HSAB	Charge density and polarizability	Predicts stable acid-base pairs

Module 6

Stereochemistry

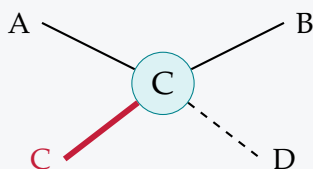
1. Stereochemistry

Stereochemistry is the study of three-dimensional arrangement of atoms in molecules and how this arrangement affects physical, chemical, and biological properties.

1.1 Representations of Three-Dimensional Structures

Three-dimensional structures are represented using wedge-dash, Fischer projection, Newman projection, and sawhorse projection.

Wedge-Dash Representation Around a Tetrahedral Carbon



solid wedge: coming out; dashed bond: going back

1.2 Structural Isomers and Stereoisomers

Structural isomers differ in connectivity, whereas stereoisomers have same connectivity but different spatial arrangement.

Type	Meaning	Example
Chain isomerism	Different carbon skeleton	n-butane and isobutane
Position isomerism	Functional group at different position	1-propanol and 2-propanol
Geometrical isomerism	Different arrangement around double bond/ring	cis-2-butene and trans-2-butene
Optical isomerism	Non-superimposable mirror images	lactic acid enantiomers

1.3 Chirality, Enantiomers and Diastereomers

A molecule is chiral if it is not superimposable on its mirror image. A common cause is a carbon attached to four different groups.

Enantiomers are non-superimposable mirror-image stereoisomers. **Diastereomers** are stereoisomers that are not mirror images.

1.4 Optical Activity and Absolute Configuration

Optically active substances rotate plane-polarized light. Absolute configuration is assigned by Cahn-Ingold-Prelog rules.

Specific Rotation

$$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$$

where α is observed rotation in degrees, l is path length in dm, and c is concentration in g mL^{-1} .

1.5 Conformational Analysis

Conformations arise by rotation about single bonds. Ethane has staggered and eclipsed conformations; staggered is more stable due to lower torsional strain.

Ethane Conformations: Staggered and Eclipsed

Staggered



Eclipsed



1.6 Isomerism in Transition Metal Compounds

Coordination compounds show geometrical, optical, linkage, ionization, and coordination isomerism. Square planar and octahedral complexes commonly show cis-trans isomerism.

Exam Points

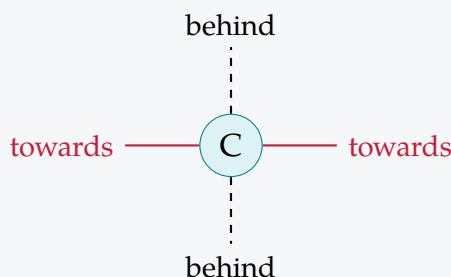
- Always distinguish structural isomerism from stereoisomerism.
- In optical isomerism, mention chirality and non-superimposable mirror images.
- CIP assignment needs priority order before clockwise/anticlockwise decision.
- For conformational analysis, compare stability using torsional strain.

2. Detailed Stereochemical Analysis

2.1 Fischer Projection Rules

Fischer projections are used mainly for molecules with chiral carbon chains. Horizontal bonds come out of the plane toward the viewer; vertical bonds go behind the plane.

Fischer Projection Convention



Rules for Fischer Projections

- Rotation by 180° gives the same molecule.
- Rotation by 90° gives a different configuration and is not allowed.
- Interchanging any two groups once inverts configuration.
- Interchanging two pairs of groups gives the original configuration.

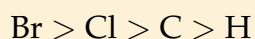
2.2 Assigning R and S Configuration

Cahn-Ingold-Prelog priority rules:

1. Atom directly attached with higher atomic number gets higher priority.
2. If first atoms are same, compare the next set of attached atoms.
3. Multiple bonds are treated as equivalent single bonds to duplicate atoms.
4. View the molecule with the lowest priority group pointing away.
5. Clockwise order $1 \rightarrow 2 \rightarrow 3$ is *R*; anticlockwise is *S*.

Worked Method: R/S Assignment

For a chiral carbon attached to Br, Cl, CH_3 , and H:



Place H away from the viewer. If the order $\text{Br} \rightarrow \text{Cl} \rightarrow \text{CH}_3$ is clockwise, the configuration is *R*; if anticlockwise, it is *S*.

2.3 Geometrical Isomerism: E/Z and Cis/Trans

Geometrical isomerism occurs due to restricted rotation around a double bond or within rings. Cis/trans notation is used for simple cases; E/Z notation is used when each double-bond carbon has two different groups.

Notation	Meaning	Use case
Cis	Similar groups on same side	Simple alkenes or rings
Trans	Similar groups on opposite sides	Simple alkenes or rings
E	Higher-priority groups on opposite sides	General alkenes by CIP rule
Z	Higher-priority groups on same side	General alkenes by CIP rule

2.4 Meso Compounds

A meso compound contains chiral centres but is optically inactive due to internal symmetry. It is superimposable on its mirror image because optical rotations of two halves cancel internally.

Meso compound: An achiral molecule with two or more stereocentres and an internal plane of symmetry.

2.5 Conformational Analysis of Butane

Butane has anti, gauche, and eclipsed conformations. Anti is most stable because bulky methyl groups are far apart.

Conformation	Dihedral angle between methyl groups	Relative stability
Anti staggered	180°	Most stable
Gauche staggered	60°	Less stable due to steric repulsion
Eclipsed	0° or 120°	Least stable due to torsional strain

Specific Rotation Numerical

If observed rotation $\alpha = +6^\circ$, path length $l = 2$ dm, and concentration $c = 0.5$ g mL⁻¹:

$$[\alpha] = \frac{\alpha}{lc} = \frac{6}{2 \times 0.5} = +6^\circ$$

2.6 Coordination Compound Stereoisomerism

Square planar complexes of type MA₂B₂ show cis-trans isomerism. Octahedral complexes may show cis-trans, fac-mer, and optical isomerism depending on ligand arrangement.

Complex type	Possible isomerism	Example pattern
Square planar MA ₂ B ₂	Cis-trans	[Pt(NH ₃) ₂ Cl ₂]
Octahedral MA ₄ B ₂	Cis-trans	Two B ligands adjacent or opposite
Octahedral MA ₃ B ₃	fac-mer	Three same ligands on one face or meridian

Stereochemistry Answer Checklist

1. Identify whether connectivity changes or only spatial arrangement changes.
2. Check for chirality, plane of symmetry, and mirror-image relation.
3. Use CIP rules for R/S or E/Z configuration.
4. Mention optical activity only after checking symmetry.

2.7 Number of Stereoisomers

For a molecule with n independent chiral centres, the maximum number of stereoisomers is:

$$2^n$$

If the molecule has internal symmetry, the actual number is less due to meso forms.

Solved Example: Number of Stereoisomers

A molecule has two chiral centres and no plane of symmetry.

$$\text{Maximum stereoisomers} = 2^2 = 4$$

These are arranged as two pairs of enantiomers. If a plane of symmetry exists, one or more meso forms reduce the total count.

2.8 Resolution of Racemic Mixtures

A racemic mixture contains equal amounts of two enantiomers and is optically inactive. Resolution means separation of enantiomers.

Method	Principle	Use
Mechanical separation	Physical separation of crystals	Rare; possible if crystals differ visibly
Chemical resolution	Formation of diastereomeric salts	Common laboratory method
Biochemical resolution	Enzyme reacts with one enantiomer faster	High selectivity
Chromatographic resolution	Chiral stationary phase separates enantiomers	Analytical and preparative use

2.9 Optical Purity and Enantiomeric Excess

Optical purity compares observed rotation with rotation of the pure enantiomer.

Optical Purity

$$\text{Optical purity} = \frac{\alpha_{\text{observed}}}{\alpha_{\text{pure}}} \times 100\%$$

For a two-enantiomer mixture, optical purity is equal to enantiomeric excess.

Solved Example: Enantiomeric Excess

If pure enantiomer has specific rotation $+40^\circ$ and mixture shows $+20^\circ$:

$$\text{optical purity} = \frac{20}{40} \times 100 = 50\%$$

So enantiomeric excess is 50%. The mixture contains 75% major enantiomer and 25% minor enantiomer.

2.10 Sequence Rules for Double Bonds: E/Z Worked Method

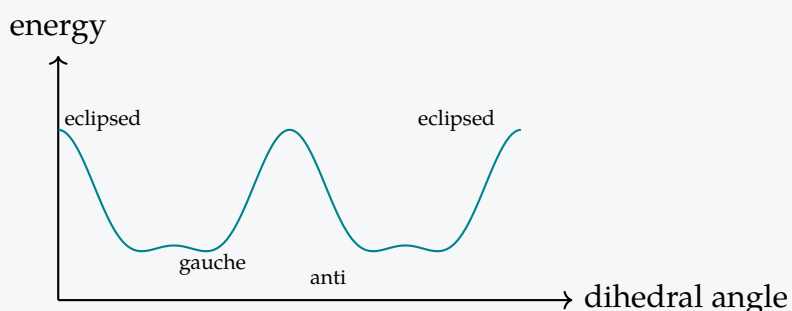
For each double-bond carbon, assign priority to its two substituents using CIP rules. If the two higher-priority groups are on the same side, the alkene is *Z*; if opposite, it is *E*.

Worked Method: E/Z Assignment

For $\text{CH}_3\text{CH}=\text{CHCl}$, compare substituents on each double-bond carbon. On the left carbon, CH_3 has higher priority than H. On the right carbon, Cl has higher priority than H. If CH_3 and Cl are on opposite sides, the configuration is *E*; if same side, it is *Z*.

2.11 Energy Profile of Butane Rotation

Rotation about the central C–C bond of butane gives different conformations. Anti is most stable; fully eclipsed is least stable.

Qualitative Energy Profile for Butane Rotation**2.12 Stereochemistry in Biological and Drug Action**

Many drug molecules are chiral, and different enantiomers can show different biological activity because enzymes and receptors are chiral environments.

Point	Importance
Different receptor fit	One enantiomer may bind better than the other
Different metabolism	Enzymes may break down enantiomers at different rates
Different toxicity	One enantiomer may be useful while the other causes side effects
Quality control	Optical purity matters in pharmaceutical synthesis

Previous-Year Style Questions

1. Explain chirality, enantiomers and diastereomers with examples.
2. State CIP rules and assign R/S configuration for a given molecule.
3. Explain optical activity, specific rotation and racemic mixture.
4. Compare conformations of ethane and butane.
5. Discuss geometrical and optical isomerism in coordination compounds.

Quick Revision — Module 6

Stereochemistry | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$[\alpha]_{\lambda}^T = \frac{\alpha}{lc}$	Specific rotation of optically active compound.
$\text{enantiomeric excess} = \frac{ R - S }{R + S} \times 100\%$	Optical purity of a mixture.
$\Delta E = E_{\text{eclipsed}} - E_{\text{staggered}}$	Torsional energy barrier in conformational analysis.
CIP priority	Higher atomic number gets higher priority.

One-Line Definitions

- **Stereoisomers:** Same connectivity but different spatial arrangement.
- **Chiral molecule:** Molecule not superimposable on its mirror image.
- **Enantiomers:** Non-superimposable mirror-image isomers.
- **Diastereomers:** Stereoisomers that are not mirror images.
- **Optical activity:** Ability to rotate plane-polarized light.
- **Conformation:** Spatial arrangement formed by rotation about single bond.
- **Racemic mixture:** Equimolar mixture of two enantiomers.

Frequently Asked / Viva Questions

Q1. What is stereochemistry?

Study of three-dimensional arrangement of atoms in molecules.

Q2. What is chirality?

Non-superimposability of a molecule on its mirror image.

Q3. Define enantiomers.

Enantiomers are non-superimposable mirror-image stereoisomers.

Q4. Define diastereomers.

Diastereomers are stereoisomers that are not mirror images.

Q5. What is optical activity?

Rotation of plane-polarized light by a chiral substance.

Q6. What is a racemic mixture?

Frequently Asked / Viva Questions (contd.)

A 1:1 mixture of enantiomers having zero net optical rotation.

Q7. What does R/S configuration indicate?

It gives absolute three-dimensional arrangement around a chiral centre.

Q8. Why is staggered ethane more stable than eclipsed ethane?

Staggered conformation has minimum torsional repulsion.

Q9. Name isomerism in coordination compounds.

Geometrical, optical, linkage, ionization, and coordination isomerism.

Q10. What is geometrical isomerism?

Isomerism due to restricted rotation causing cis-trans or fac-mer forms.

Summary Reference Table

Pair	Relation	Key difference
Structural isomers	Different connectivity	Different bonding sequence
Enantiomers	Mirror images	Equal and opposite optical rotation
Diastereomers	Not mirror images	Different physical properties
Conformers	Rotation about single bond	Interconvert without bond breaking

Module 7

Organic Reactions and Drug Molecule Synthesis

1. Organic Reactions and Drug Molecule Synthesis

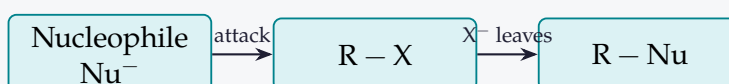
Organic reaction mechanism explains the step-by-step breaking and making of bonds during conversion of reactants into products. It identifies intermediates, reagents, and electron movement.

1.1 Substitution Reactions

In substitution, one atom or group is replaced by another. Nucleophilic substitution is common in alkyl halides.

Type	Main feature	Favoured by
S _N 1	Two-step, carbocation intermediate	Tertiary substrate, polar protic solvent
S _N 2	One-step backside attack	Methyl/primary substrate, strong nucleophile

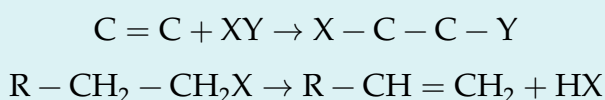
SN2 Backside Attack Concept



1.2 Addition and Elimination Reactions

Addition reactions add atoms across multiple bonds. Elimination reactions remove atoms from adjacent carbons to form a double bond.

Typical Patterns



1.3 Oxidation, Reduction, Cyclization and Ring Opening

Oxidation increases C-O, C-N, or C-X bonds or decreases C-H bonds. Reduction increases C-H bonds or removes electronegative atoms. Cyclization forms a ring; ring opening

converts cyclic compounds into open-chain products.

Reaction	Meaning	Example idea
Oxidation	Gain of oxygen or loss of hydrogen	Alcohol to aldehyde/acid
Reduction	Gain of hydrogen or loss of oxygen	Carbonyl to alcohol
Cyclization	Ring formation	Intramolecular attack
Ring opening	Cleavage of a ring bond	Epoxide hydrolysis

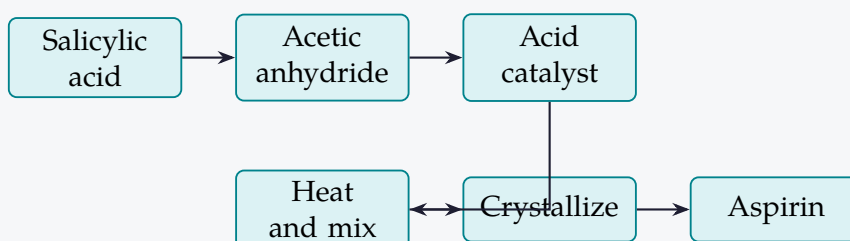
1.4 Synthesis of a Common Drug Molecule: Aspirin

Aspirin (acetylsalicylic acid) is prepared by acetylation of salicylic acid using acetic anhydride in acidic medium.

Aspirin Synthesis Reaction



Flow Diagram for Aspirin Preparation



1.5 Mechanistic Exam Language

For mechanism-based answers, identify electrophile, nucleophile, leaving group, intermediate, and product. Use curved-arrow reasoning in words even when detailed curved arrows are not drawn.

Exam Points

- Compare S_N1 and S_N2 using step, intermediate, rate law, and stereochemistry.
- For addition and elimination, write the general reaction pattern.
- For oxidation/reduction, state change in C-H and C-O bonds.
- For aspirin synthesis, mention salicylic acid, acetic anhydride, acid catalyst, and acetylation.

2. Detailed Organic Reaction Notes and Synthesis Planning

2.1 Reaction Intermediates

Organic reactions often proceed through reactive intermediates. Stability of intermediates decides product distribution and rate.

Intermediate	Nature	Stability trend
Carbocation	Electron-deficient carbon with positive charge	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$
Carbanion	Electron-rich carbon with negative charge	Stabilized by electron-withdrawing groups
Free radical	Species with unpaired electron	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3\cdot$
Carbene	Neutral divalent carbon species	Highly reactive; addition to alkenes

2.2 Detailed Comparison of $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$

Point	$\text{S}_\text{N}1$	$\text{S}_\text{N}2$
Mechanism	Two-step	One-step concerted
Rate law	$k[\text{RX}]$	$k[\text{RX}][\text{Nu}^-]$
Intermediate	Carbocation	No intermediate
Substrate	$3^\circ > 2^\circ > 1^\circ$	methyl $> 1^\circ > 2^\circ$
Solvent	Polar protic	Polar aprotic
Stereochemistry	Racemization possible	Inversion of configuration

Solved Concept: Choosing $\text{S}_\text{N}1$ or $\text{S}_\text{N}2$

Tertiary butyl chloride reacts with water mainly by $\text{S}_\text{N}1$ because a stable tertiary carbocation can form and water is a weak nucleophile in polar protic medium. Methyl bromide reacts with OH^- mainly by $\text{S}_\text{N}2$ because backside attack is unhindered.

2.3 Electrophilic Addition to Alkenes

Alkenes undergo electrophilic addition because the pi bond is electron rich. Addition of HX follows Markovnikov rule in normal ionic addition.

Markovnikov rule: In addition of HX to an unsymmetrical alkene, hydrogen attaches to the carbon already having more hydrogen atoms, and X attaches to the more substituted carbon.

Addition of HBr to Propene



The major product is 2-bromopropane due to formation of the more stable secondary carbocation.

2.4 Elimination: E1 and E2

Elimination reactions form alkenes. E1 proceeds through carbocation; E2 occurs in one step with base removing beta-hydrogen as leaving group departs.

Point	E1	E2
Steps	Two-step	One-step
Intermediate	Carbocation	None
Base strength	Weak base possible	Strong base required
Substrate	Tertiary favoured	Primary, secondary, tertiary possible
Stereochemical need	No strict anti requirement	Anti-periplanar H and leaving group preferred

2.5 Oxidation and Reduction Reagents

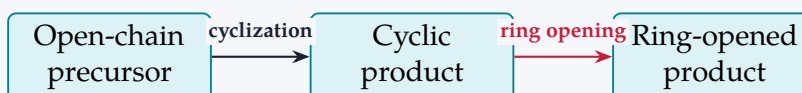
Remembering reagent function is more useful than memorizing long mechanisms for basic engineering chemistry.

Reagent	Type	Common use
KMnO ₄	Strong oxidizing agent	Alkene oxidation, alcohol oxidation
K ₂ Cr ₂ O ₇ /H ⁺	Oxidizing agent	Primary alcohol to acid, secondary alcohol to ketone
LiAlH ₄	Strong reducing agent	Carbonyl and acid derivatives to alcohols
NaBH ₄	Mild reducing agent	Aldehyde/ketone to alcohol
H ₂ /Ni	Catalytic reduction	Alkene to alkane

2.6 Cyclization and Ring Opening

Cyclization is favoured when the reacting ends of a molecule can form a stable five- or six-membered ring. Ring opening is important for strained rings like epoxides.

Cyclization and Ring Opening Concept



2.7 Aspirin Synthesis: Mechanism and Purification

In aspirin preparation, salicylic acid provides the phenolic –OH group and acetic anhydride provides the acetyl group. Acid catalyst increases electrophilicity of acetic anhydride. The product is purified by recrystallization.

1. Protonation of acetic anhydride increases electrophilic character of carbonyl carbon.
2. Phenolic oxygen of salicylic acid attacks the carbonyl carbon.
3. Acetate group leaves and deprotonation gives aspirin.

4. Cold water helps crystallize aspirin because aspirin is less soluble in cold water.

Aspirin Yield Calculation

If theoretical yield is 5.00 g and actual yield is 3.80 g:

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$\% \text{ yield} = \frac{3.80}{5.00} \times 100 = 76\%$$

Organic Reaction Answer Checklist

1. Identify substrate, reagent, nucleophile/electrophile, and leaving group.
2. Classify the reaction as substitution, addition, elimination, oxidation, reduction, cyclization, or ring opening.
3. Write rate law or regioselectivity rule where applicable.
4. Mention major product and reason for its formation.
5. For drug synthesis, include reactants, catalyst, reaction type, purification, and yield idea.

2.8 Reaction Map for Basic Organic Transformations

Most exam questions ask how one functional group changes into another. The following map gives the common first-year transformations.

Conversion	Typical reagent	Reaction class
Alkene to alkane	H ₂ /Ni	Reduction / hydrogenation
Alkene to alcohol	H ₂ O/H ⁺	Addition
Alcohol to alkene	conc. H ₂ SO ₄ , heat	Elimination / dehydration
Primary alcohol to acid	KMnO ₄ or K ₂ Cr ₂ O ₇ /H ⁺	Oxidation
Aldehyde to alcohol	NaBH ₄	Reduction
Alkyl halide to alcohol	aqueous KOH	Nucleophilic substitution

2.9 Curly-Arrow Logic in Words

Even when curved arrows are not drawn, the answer should describe electron flow clearly:

1. A nucleophile donates an electron pair.
2. An electrophile accepts an electron pair.
3. A leaving group departs with the bonding electron pair.
4. Formation of a stable intermediate or transition state controls the major product.

Worked Mechanism in Words: Alcohol to Alkene

In acid-catalysed dehydration of an alcohol, the —OH group is first protonated to form a better leaving group, water. Water leaves to form a carbocation. A base removes a beta-hydrogen, and a double bond forms. The major alkene is usually the more substituted alkene according to Zaitsev rule.

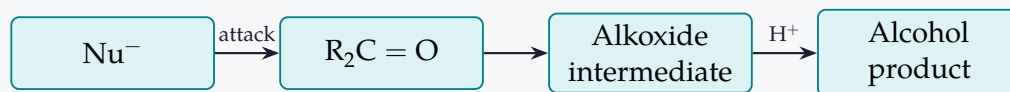
2.10 Zaitsev and Hofmann Elimination

When more than one alkene can form, regioselectivity matters.

Rule	Major product	Typical condition
Zaitsev rule	More substituted alkene	Small strong base or normal elimination
Hofmann rule	Less substituted alkene	Bulky base or quaternary ammonium elimination

2.11 Addition to Carbonyl Compounds

The carbonyl group is polar. Carbon is electrophilic and oxygen is nucleophilic/basic. Nucleophiles attack the carbonyl carbon to form addition products.

Carbonyl Addition Concept**2.12 Green Chemistry Note for Drug Synthesis**

Engineering chemistry answers may include practical synthesis quality. A good drug synthesis should be high-yielding, selective, safe, and easy to purify.

Green chemistry point	Meaning in aspirin synthesis
Atom economy	Prefer maximum incorporation of reactant atoms into product
Safer solvent	Use minimum harmful solvent during purification
Energy efficiency	Avoid unnecessary heating time
Waste reduction	Reduce excess reagent and by-product handling
Product purity	Recrystallization improves purity for medicinal use

2.13 Tests for Functional Groups

Short practical-style questions may ask how a functional group is detected.

Functional group	Test	Positive observation
Alkene	Bromine water	Decolorization
Aldehyde	Tollens' reagent	Silver mirror
Carboxylic acid	Sodium bicarbonate	Brisk effervescence of CO ₂
Phenol	Ferric chloride test	Violet or coloured complex
Alcohol	Esterification	Fruity smell of ester

Previous-Year Style Questions

1. Compare S_N1 and S_N2 mechanisms with stereochemical outcome.
2. Explain electrophilic addition to propene using Markovnikov rule.
3. Compare E1 and E2 elimination reactions.
4. Write short notes on oxidation and reduction reagents in organic chemistry.
5. Describe aspirin synthesis, purification and percentage yield calculation.

Quick Revision — Module 7

Organic Reactions and Drug Molecule Synthesis | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\text{rate}_{S_N1} = k[\text{RX}]$	First-order nucleophilic substitution.
$\text{rate}_{S_N2} = k[\text{RX}][\text{Nu}^-]$	Second-order nucleophilic substitution.
Addition pattern	$\text{C} = \text{C} + \text{XY} \rightarrow \text{X} - \text{C} - \text{C} - \text{Y}$ General addition across a multiple bond.
Elimination pattern	$\text{R} - \text{CH}_2 - \text{CH}_2\text{X} \rightarrow \text{R} - \text{CH} = \text{CH}_2 + \text{HX}$ Removal of HX to form an alkene.
Aspirin synthesis	Salicylic acid + Acetic anhydride $\rightarrow$ Aspirin + Acetic acid Acetylation reaction for aspirin preparation.

One-Line Definitions

- **Nucleophile:** Electron-rich species that attacks electron-poor centre.
- **Electrophile:** Electron-deficient species that accepts electron pair.
- **Leaving group:** Group that departs with electron pair.
- **Substitution:** Replacement of one group by another.
- **Addition:** Addition of atoms across a multiple bond.
- **Elimination:** Removal of groups to form a multiple bond.
- **Acetylation:** Introduction of acetyl group, $\text{CH}_3\text{CO}-$.

Frequently Asked / Viva Questions

Q1. What is an organic reaction mechanism?

It is the stepwise description of bond breaking and bond formation.

Q2. Define nucleophile.

A nucleophile is an electron-rich species that donates an electron pair.

Q3. Define electrophile.

An electrophile is an electron-deficient species that accepts an electron pair.

Q4. Compare S_N1 and S_N2 briefly.

S_N1 is two-step and forms carbocation; S_N2 is one-step backside attack.

Frequently Asked / Viva Questions (contd.)

Q5. What is an addition reaction?

A reaction in which atoms or groups add across a double or triple bond.

Q6. What is an elimination reaction?

A reaction in which small molecules are removed to form a multiple bond.

Q7. How is oxidation recognized in organic chemistry?

Increase in C-O/C-X bonds or decrease in C-H bonds.

Q8. How is reduction recognized in organic chemistry?

Increase in C-H bonds or decrease in C-O/C-X bonds.

Q9. What is cyclization?

Intramolecular reaction leading to ring formation.

Q10. Name reactants used for aspirin synthesis.

Salicylic acid and acetic anhydride in presence of acid catalyst.

Summary Reference Table

Reaction type	Core change	Exam clue
Substitution	Group replacement	Leaving group present
Addition	Break pi bond, add groups	Alkene/alkyne reactant
Elimination	Remove small molecule	Alkene product
Oxidation/reduction	Change in C-O and C-H bonds	Reagent decides direction
Drug synthesis	Planned sequence	Reactants, catalyst, product